

NOTE ON THE SMITH-WILSON INTEREST RATE CURVE

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ABSTRACT. Since the entry into force of Solvency II as of 1 January 2016, all European insurance companies concerned have to use the Smith-Wilson interest rate curve to determine the value of their insurance obligations and thus of a substantial part of their balance sheet. Although Smith & Wilson (2000) introduce the underlying discount curve $\underline{P}(t)$ as the sum of a ‘long-term’ discount curve $e^{-f_\infty t}$ and a linear combination of the so-called Wilson function $W(t, u)$ evaluated at different payment dates u_j , that is,

$$\underline{P}(t) = e^{-f_\infty t} + \sum_j \beta_j W(t, u_j),$$

a mathematically sound derivation of its shape is missing in the literature. The aim of this paper is to close this gap. To this end, we reformulate the infinite-dimensional optimization problem stated in Smith & Wilson (2000) within an analytically rigorous framework. We prove that it has a unique minimizer and explicitly derive the formula displayed above. In doing so, we show that $W(t, u)$ is in fact the kernel function of a particular reproducing kernel Hilbert space, which is the key result to fully understanding the shape of $\underline{P}(t)$.

1. INTRODUCTION

The centerpiece of valuation under Solvency II is the relevant term structure of risk-free interest rates (see Article 77(2) of Directive 2009/138/EC), a description of which can be found in an official EIOPA paper with the title ‘Technical documentation of the methodology to derive EIOPA’s risk-free interest rate term structures’. By referring to an unpublished note by Smith & Wilson (2000), this technical documentation introduces the so-called Smith-Wilson method, the main feature of which is to allow for the ultimate forward rate $f_\infty := \lim_{t \rightarrow \infty} f(t)$ to be fixed as an exogenous parameter. Furthermore, the discount curve $\underline{P}(t)$ can be written as the sum of a ‘long-term’ discount curve $e^{-f_\infty t}$ and a linear combination of functions $W(t, u_j)$, that is,

$$(1) \quad \underline{P}(t) = e^{-f_\infty t} + \sum_j \beta_j W(t, u_j),$$

where the underlying risk-free financial instruments make payments at times u_j . Although all formulas necessary for implementation are provided, a derivation of $W(t, u)$, the so-called Wilson function, and consequently of the discount curve is missing.

This is also the case for the original paper by Smith & Wilson (2000). There the authors lay down their idea of finding a discount curve by making the implied forward curve as ‘flat’ and as ‘smooth’ as possible, whilst matching prices of prespecified bonds. But then, without any explanation about its derivation, they present a solution of the related optimization problem and, only by drawing on its explicit form, are able to show that it is indeed the only solution.

In a recent paper, de Kort & Vellekoop (2016) extend the Smith-Wilson method by considering f_∞ as a free parameter to be estimated from market data. Their results are based on an independent proof of the Smith-Wilson optimization problem, which has certain parallels to the proof presented in this paper but is unsatisfactory insofar as the authors make repeated use of the assumption that

$$(2) \quad \left. \frac{d^2}{dt^2} \right|_{t=0} [e^{f_\infty t} P(t)] = 0$$

at crucial points in their argumentation. Imposing this assumption as an a priori restriction is problematic in two respects: First, discount curves, such as the discount curves belonging to the Nelson-Siegel family, do not, in general, have the property stated in (2). Second, (2) is a consequence of solving the Smith-Wilson optimization problem rather than an assumption needed to solve it.

In this paper, we explain how to find a solution to the Smith-Wilson optimization problem within a mathematically sound framework. We first prove that the objective function has a unique minimizer $\underline{P}(t)$ and, once knowing this, explicitly derive its shape. In doing so, it turns out that the Wilson function $W(t, u)$ is in fact the kernel function of a particular reproducing kernel Hilbert space.

2. DEFINITIONS AND NOTATION

In this section, we introduce some basic concepts that are used throughout the rest of the paper. We start with the term structure of zero-coupon bond prices.

2.1. The term structure of zero-coupon bond prices. For $s, t \in [0, \infty)$ such that $t \geq s$, we denote by $P(s, t)$ the time s value of a zero-coupon bond paying out one unit at maturity t . The function $t \mapsto P(s, t)$ is the term structure of zero-coupon bond prices at time s . The continuously compounded spot rate $R(s, t)$ with maturity t at time s is given by

$$(3) \quad R(s, t) = -\frac{\log P(s, t)}{t - s};$$

and the forward rate $f(s, t)$ with maturity t at time s is given by

$$(4) \quad f(s, t) = -\frac{\partial \log P(s, t)}{\partial t}.$$

Letting $s = 0$, define the discount curve $t \mapsto P(t) := P(0, t)$ (notice that $P(t)$ is the price today of a payout of one unit at time t), the spot curve $t \mapsto R(t) := R(0, t)$, and the forward curve $t \mapsto f(t) := f(0, t)$.

2.2. Fréchet differentiability. Let E and F be Banach spaces and let $Q : U \rightarrow F$ be a mapping on some open set $U \subseteq E$. Let $u_0 \in U$. If there exists a bounded linear mapping $DQ(u_0) : E \rightarrow F$ such that

$$(5) \quad Q(u_0 + h) - Q(u_0) - DQ(u_0)(h) = o(h) \text{ as } h \rightarrow 0,$$

then Q is said to be Fréchet differentiable at u_0 . The mapping $DQ(u_0)$ is unique and is called the Fréchet derivative of Q at u_0 .

If Q is Fréchet differentiable at u_0 , then, for any $h \in E$, the directional derivative of Q at u_0 into the direction $h \in E$, given by

$$(6) \quad d_h Q(u_0) := \lim_{t \rightarrow 0} \frac{Q(u_0 + th) - Q(u_0)}{t},$$

exists and is equal to $DQ(u_0)(h)$.

For $g : \Omega \subseteq \mathbb{R} \rightarrow \mathbb{R}$, Ω some open subset of $\mathbb{R}$, the existence of the ordinary derivative at $u_0 \in \Omega$, i.e. of $d_1 g(u_0)$, already implies Fréchet differentiability at u_0 , so that the concepts of ordinary and Fréchet differentiability coincide for real-valued functions.

For $g : \bar{\Omega} \rightarrow \mathbb{R}$, $\Omega \subseteq \mathbb{R}$ an open interval, the derivative of order m at $u_0 \in \bar{\Omega}$ (the one-sided derivative in case u_0 belongs to the boundary) is denoted by $\frac{d^m}{dt^m} g(u_0)$ or $g^{(m)}(u_0)$.

2.3. Weak differentiability and Sobolev spaces. Let Ω be a non-empty open interval in $\mathbb{R}$. A Borel measurable function $g : \Omega \rightarrow \mathbb{R}$ is said to be weakly differentiable if there exists a function $z : \Omega \rightarrow \mathbb{R}$ such that

$$(7) \quad \int_{\Omega} z(t) \varphi(t) dt = - \int_{\Omega} g(t) \varphi^{(1)}(t) dt$$

holds for all $\varphi \in C^\infty(\Omega)$ with compact support in Ω . The function z is unique up to equality almost everywhere on Ω , so that we can define the weak derivative $D_w g$ of g as the equivalence class of z with respect to the relation of being equal almost everywhere. If g is differentiable with locally Lebesgue integrable derivative, then g is weakly differentiable with weak derivative $z = \frac{d}{dt} g(t)$.

For functions $g, h : \Omega \rightarrow \mathbb{R}$, let

$$(8) \quad \begin{aligned} \langle g|h \rangle_{1,2} : &= \langle g|h \rangle_2 + \langle D_w g | D_w h \rangle_2 \\ &= \int_{\Omega} g(t) h(t) dt + \int_{\Omega} D_w g(t) D_w h(t) dt \end{aligned}$$

and $\|g\|_{1,2} := \sqrt{\langle g|g \rangle_{1,2}}$. Here, $\langle \cdot | \cdot \rangle_2$ is the usual (indefinite) inner product on the space $\mathcal{L}^2(\Omega)$ of Borel measurable functions $g : \Omega \rightarrow \mathbb{R}$ that satisfy $\int_{\Omega} g(t)^2 dt < \infty$. We define $W^{1,2}(\Omega)$ as the space of equivalence classes $[g]$, with respect to the relation of being equal almost everywhere, of all Borel measurable functions $g : \Omega \rightarrow \mathbb{R}$ such that $\|g\|_{1,2}$ is finite. [More generally, a definition of the spaces $W^{k,p}(\Omega)$, where k is a non-negative integer and $1 \leq p < \infty$, is provided in Section 3.1 of the classical book by Adams, 1975.]

It follows from Part II of Theorem 5.4 [Sobolev imbedding theorem] in Adams (1975) (with $m = n = 1, p = 2, j = 0$) that each equivalence class in $W^{1,2}(\Omega)$ contains exactly one bounded, uniformly continuous function on Ω . We consequently define the Sobolev space of order 1 as

$$(9) \quad W^{1,2}(\Omega) := \{g : \Omega \rightarrow \mathbb{R} : g \text{ is bounded, uniformly continuous and } [g] \in W^{1,2}(\Omega)\}$$

and we note that $W^{1,2}(\Omega)$ is a Hilbert space. For the convenience of the reader, a short proof of this fact can be found in Proposition A.3 in the appendix.

As discount curves are naturally defined on $[0, \infty)$, we need an extension of the definition of Sobolev spaces to non-open intervals in $\mathbb{R}$. To this end, observe that since the elements of $W^{1,2}(\Omega)$ are bounded, uniformly continuous functions on Ω , each $g \in W^{1,2}(\Omega)$ can be uniquely extended to a function $\bar{g} : \bar{\Omega} \rightarrow \mathbb{R}$ on the closure $\bar{\Omega}$ of Ω , which again is bounded and uniformly continuous. Thus, for Ω a non-empty open interval of $\mathbb{R}$, the Sobolev space of order 1 on $\bar{\Omega}$ is defined as

$$(10) \quad W^{1,2}(\bar{\Omega}) := \{\bar{g} : \bar{\Omega} \rightarrow \mathbb{R} : g \in W^{1,2}(\Omega)\}$$

with $\langle \bar{g} | \bar{h} \rangle_{1,2} := \langle g | h \rangle_{1,2}$. Some useful properties of $W^{1,2}(\bar{\Omega})$, including the case $W^{1,2}([0, \infty))$, are collected in the appendix.

3. THE SMITH-WILSON OPTIMIZATION PROBLEM

In the following, the financial market is considered to be deep, liquid, and transparent up to time T , which is called last liquid point. Note that for the purpose of fitting the relevant risk-free interest rate term structure under Solvency II, the last liquid point is fixed in EIOPA-BoS-15/035 (2016). Depending on the currency, it can range from 10 years, as for the Croatian kuna, up to 50 years, as for the pound sterling; and it is set equal to 20 years for the euro (see EIOPA-BoS-15/035, 2016).

Within this market, let $B_1, \dots, B_n$ be n risk-free financial instruments with corresponding payments c_{ji} , $i = 1, \dots, n$, at times u_j , $j = 1, \dots, J$. Let $B_1, \dots, B_n$ be chosen in such a way that $u_J = T$; and denote by $v_1, \dots, v_n$ their respective market values.

In their paper, Smith & Wilson (2000) are looking for a discount curve $P(t)$, $t \geq 0$, such that, for $i = 1, \dots, n$,

$$(11) \quad \sum_{j=1}^J c_{ji} P(u_j) = v_i$$

is satisfied and

$$(12) \quad \alpha^{-1} \int_0^\infty \left(\frac{d}{dt} [e^{f_\infty t} P(t)] \right)^2 dt + \alpha^{-3} \int_0^\infty \left(\frac{d^2}{dt^2} [e^{f_\infty t} P(t)] \right)^2 dt$$

is minimal. In other words, the discount curve is defined as the solution of the optimization problem

$$(13) \quad \begin{aligned} & \text{minimize} \quad \alpha^{-1} \int_0^\infty \left(\frac{d}{dt} [e^{f_\infty t} P(t)] \right)^2 dt + \alpha^{-3} \int_0^\infty \left(\frac{d^2}{dt^2} [e^{f_\infty t} P(t)] \right)^2 dt \\ & \text{over} \quad P(t), t \geq 0, \end{aligned}$$

under the constraints

$$(14) \quad \begin{aligned} & P(0) = 1 \\ & \sum_{j=1}^J c_{j1} P(u_j) = v_1 \\ & \quad \vdots \\ & \sum_{j=1}^J c_{jn} P(u_j) = v_n. \end{aligned}$$

The choice of the objective function can be motivated as follows (see Smith & Wilson, 2000): Calculating the derivatives, the square-root of the first integrand is equal to

$$(15) \quad \left| f_\infty e^{f_\infty t} P(t) + e^{f_\infty t} P(t) \frac{d}{dt} \log P(t) \right| = |(f_\infty - f(t)) e^{f_\infty t} P(t)|$$

and that of the second integrand is equal to

$$(16) \quad \left| \left(-\frac{d}{dt} f(t) + (f_\infty - f(t))^2 \right) e^{f_\infty t} P(t) \right|.$$

Thus, if the forward rate $f(t)$ is ‘flat’ in the sense that $f(t)$ is close to the ultimate forward rate f_∞ and if $f(t)$ is ‘smooth’ in the sense that $\frac{d}{dt} f(t)$ is small, then $P(t)$ lies close to the desired discount curve.

As the powers of $\alpha > 0$ appearing in (13) are not equal, α is a parameter determining the speed of convergence of the forward rate $f(t)$ to the ultimate forward rate

$$(17) \quad f_\infty = \lim_{t \rightarrow \infty} f(t) = - \lim_{t \rightarrow \infty} \frac{d}{dt} \log P(t)$$

(see the remarks following the proof of Theorem 5.1). Note that f_∞ is laid down by European regulation (see EIOPA-BoS-15/035, 2016) and, in particular, is set equal to 4.2% for the euro.

3.1. A model space for discount curves. We require all possible discount curves to be elements of the space of functions

$$(18) \quad \mathcal{H}_1 := \left\{ P : [0, \infty) \rightarrow \mathbb{R} : P(0) = 1, \frac{d}{dt} [e^{f_\infty t} P(t)] \in \mathcal{W}^{1,2}([0, \infty)) \right\},$$

which is slightly less than requiring the integrals appearing in (13) to exist. We thus model the discount curves under slightly weaker assumptions than Smith & Wilson (2000) do.

In the sequel, we shall need the decomposition of any $P \in \mathbf{H}_1$ into the ‘long-term’ discount curve $e^{-f\infty t}$ and the remainder $g = P - e^{-f\infty t}$ lying in

$$(19) \quad \mathbf{H}_0 := \left\{ g : [0, \infty) \rightarrow \mathbb{R} : g(0) = 0, \frac{d}{dt} [e^{f\infty t} g(t)] \in \mathbf{W}^{1,2}([0, \infty)) \right\}.$$

Clearly, $\mathbf{H}_0$ is a linear space. Hence, $\mathbf{H}_1 = e^{-f\infty t} + \mathbf{H}_0$ is an affine space.

For functions $g, h : [0, \infty) \rightarrow \mathbb{R}$, define

$$(20) \quad \begin{aligned} \langle g|h \rangle : &= \alpha^{-1} \int_{(0, \infty)} \frac{d}{dt} [e^{f\infty t} g(t)] \frac{d}{dt} [e^{f\infty t} h(t)] dt \\ &+ \alpha^{-3} \int_{(0, \infty)} D_w \frac{d}{dt} [e^{f\infty t} g(t)] D_w \frac{d}{dt} [e^{f\infty t} h(t)] dt. \end{aligned}$$

We claim that $\mathbf{H}_0$, when equipped with the inner product $\langle g|h \rangle$ and norm $\|g\| := \sqrt{\langle g|g \rangle}$ (if $\|g\| = 0$, then $\frac{d}{dt} [e^{f\infty t} g(t)] = 0$; hence $e^{f\infty t} g = 0$ as it has to be 0 at $t = 0$; hence $g = 0$, which shows the definiteness of $\|\cdot\|$) is a Hilbert space for any $\alpha > 0$. To see this, let g_n be an arbitrary Cauchy sequence in $\mathbf{H}_0$. By Part 1 of Proposition A.4 in the appendix, $\mathbf{W}^{1,2}([0, \infty))$ is complete, so that $\frac{d}{dt} [e^{f\infty t} g_n(t)]$ converges to some bounded, uniformly continuous function $w \in \mathbf{W}^{1,2}([0, \infty))$. Letting

$$(21) \quad g(t) := e^{-f\infty t} \int_{[0, t]} w(s) ds,$$

we have $g \in \mathbf{H}_0$. Since

$$(22) \quad \|g_n - g\| \leq \sqrt{\max(\alpha^{-1}, \alpha^{-3})} \left\| \frac{d}{dt} [e^{f\infty t} g_n(t)] - w \right\|_{1,2},$$

we see that g_n converges to g in $\mathbf{H}_0$.

4. THE WILSON FUNCTION

The central idea leading to the solution of the Smith-Wilson optimization problem stated in (13) is to find explicit Riesz representer of the point evaluation functionals δ_u , $\delta_u(h) = h(u)$, with u arbitrary but fixed. By definition, a Riesz representer of δ_u is an element $u^* \in \mathbf{H}_0$ such that $\langle u^*|h \rangle = \delta_u(h)$ for all $h \in \mathbf{H}_0$. As is stated in Theorem 4.1 below, these representers exist in $\mathbf{H}_0$ and do have closed-formula expressions.

Theorem 4.1. *Let $u \in [0, \infty)$ be arbitrary but fixed. Then the corresponding point evaluation functional $\delta_u : \mathbf{H}_0 \rightarrow \mathbb{R}$, given by $\delta_u(h) = h(u)$ for $h \in \mathbf{H}_0$, has a representer $u^* \in \mathbf{H}_0$; that is: $\langle u^*|h \rangle = h(u)$ holds true for all $h \in \mathbf{H}_0$. Furthermore, u^* has the form*

$$(23) \quad u^*(t) = e^{-f\infty(t+u)} \begin{cases} \alpha t - e^{-\alpha u} \sinh(\alpha t) & \text{for } 0 \leq t \leq u; \\ \alpha u - e^{-\alpha t} \sinh(\alpha u) & \text{for } t \geq u. \end{cases}$$

Proof. Let $u \in [0, \infty)$ be fixed. We first show that $\delta_u : \mathbf{H}_0 \rightarrow \mathbb{R}$, $\delta_u(h) = h(u)$, is continuous on $\mathbf{H}_0$. To this end, let $h_n, h \in \mathbf{H}_0$ be such that $\lim_n \|h_n - h\| = 0$. With $g_n := h_n - h$, we have that

$$\begin{aligned}
|g_n(u)| &\leq |e^{f_\infty u} g_n(u)| = \left| \int_0^u \frac{d}{dt} [e^{f_\infty t} g_n(t)] dt \right| \\
&\leq \left\| \frac{d}{dt} [e^{f_\infty t} g_n(t)] \right\|_{1, [0, u]} \leq \sqrt{u} \left\| \frac{d}{dt} [e^{f_\infty t} g_n(t)] \right\|_{2, [0, u]} \leq \sqrt{\alpha u} \|g_n\|,
\end{aligned}$$

where the equality in the first line of the display above follows from the fundamental theorem of calculus. As the right-hand side converges to 0, so does the left-hand side.

It follows from the Riesz representation theorem that there is a unique element $u^* \in \mathbf{H}_0$ such that $\langle u^* | h \rangle = h(u)$ holds true for all $h \in \mathbf{H}_0$. To find u^* , assume it is C^2 on $[0, \infty)$ and smooth on the intervals $[0, u]$ and $[u, \infty)$. [It turns out that u^* does in fact have this property.] Then, by definition of the inner product in Section 3.1 and by making use of the continuity of δ_u together with Part 3 of Proposition A.4 in the appendix, this is equivalent to finding the one $u^* \in \mathbf{H}_0$ such that

$$\begin{aligned}
I + II &:= \alpha^{-1} \int_{[0, \infty)} \frac{d}{dt} [e^{f_\infty t} u^*(t)] \frac{d}{dt} [e^{f_\infty t} h(t)] dt \\
&\quad + \alpha^{-3} \int_{[0, \infty)} \frac{d^2}{dt^2} [e^{f_\infty t} u^*(t)] \frac{d^2}{dt^2} [e^{f_\infty t} h(t)] dt \\
(24) \qquad &= h(u)
\end{aligned}$$

for all $h \in \mathbf{H}_0 \cap C^\infty([0, \infty))$ with compact support. In the following, we use the abbreviations $x : [0, u] \rightarrow \mathbb{R}$, $x(t) = e^{f_\infty t} u^*(t)$; $y : [u, \infty) \rightarrow \mathbb{R}$, $y(t) = e^{f_\infty t} u^*(t)$; and $z(t) = e^{f_\infty t} h(t)$. Applying the partial integration formula in such a manner that no expression of the form $z^{(1)}$ or $z^{(2)}$ (only z itself) remains under the integral signs, we get for the first integral in (24) that

$$\begin{aligned}
\alpha I &= [x^{(1)} z]_0^u - \int_0^u x^{(2)}(t) z(t) dt \\
(25) \qquad &\quad + [y^{(1)} z]_u^\infty - \int_u^\infty y^{(2)}(t) z(t) dt.
\end{aligned}$$

For the second integral in (24) we have

$$\begin{aligned}
\alpha^3 II &= [x^{(2)} z^{(1)}]_0^u - \int_0^u x^{(3)}(t) z^{(1)}(t) dt + [y^{(2)} z^{(1)}]_u^\infty - \int_u^\infty y^{(3)}(t) z^{(1)}(t) dt \\
&= [x^{(2)} z^{(1)}]_0^u - [x^{(3)} z]_0^u + \int_0^u x^{(4)}(t) z(t) dt \\
(26) \qquad &\quad + [y^{(2)} z^{(1)}]_u^\infty - [y^{(3)} z]_u^\infty + \int_u^\infty y^{(4)}(t) z(t) dt.
\end{aligned}$$

Taking into account that $z(0) = 0$ (because $h \in \mathbf{H}_0$), that z as well as $z^{(1)}$ have compact support, which in turn implies that

$$\begin{aligned}
x^{(1)}(0)z(0) &= 0 \\
x^{(3)}(0)z(0) &= 0 \\
\lim_{t \rightarrow \infty} y^{(1)}(t)z(t) &= 0 \\
\lim_{t \rightarrow \infty} y^{(3)}(t)z(t) &= 0 \\
(27) \quad \lim_{t \rightarrow \infty} y^2(t)z^{(1)}(t) &= 0,
\end{aligned}$$

and assuming

$$\begin{aligned}
x^{(4)} - \alpha^2 x^{(2)} &= 0 \\
(28) \quad y^{(4)} - \alpha^2 y^{(2)} &= 0
\end{aligned}$$

as well as

$$\begin{aligned}
x^{(2)}(0) &= 0 \\
x^{(1)}(u) - y^{(1)}(u) &= 0 \\
x^{(2)}(u) - y^{(2)}(u) &= 0 \\
(29) \quad -\alpha^{-3} e^{f_\infty u} (x^{(3)}(u) - y^{(3)}(u)) &= 1,
\end{aligned}$$

we see that $I + II = h(u)$ for all $h \in \mathbf{H}_0 \cap C^\infty([0, \infty))$ with compact support. In other words, we have to solve the system of ODEs

$$\begin{aligned}
x^{(4)} - \alpha^2 x^{(2)} &= 0 \quad \text{on } [0, u] \\
(30) \quad y^{(4)} - \alpha^2 y^{(2)} &= 0 \quad \text{on } [u, \infty)
\end{aligned}$$

with boundary conditions (29).

To this end, note that the characteristic equation of $x^{(4)} - \alpha^2 x^{(2)} = 0$ as well as the one of $y^{(4)} - \alpha^2 y^{(2)} = 0$ is given by $v^4 - \alpha^2 v^2 = 0$ and has the roots $v_1 = 0$, $v_2 = 0$, $v_3 = \alpha$, $v_4 = -\alpha$. A basis of solutions is therefore given by

$$(31) \quad x(t) = c_1 + c_2 t + c_3 e^{\alpha t} + c_4 e^{-\alpha t} \quad \text{and} \quad y(t) = d_1 + d_2 t + d_3 e^{\alpha t} + d_4 e^{-\alpha t},$$

respectively, with derivatives

$$\begin{aligned}
x^{(1)}(t) &= c_2 + c_3 \alpha e^{\alpha t} - c_4 \alpha e^{-\alpha t} & y^{(1)}(t) &= d_2 + d_3 \alpha e^{\alpha t} - d_4 \alpha e^{-\alpha t} \\
x^{(2)}(t) &= c_3 \alpha^2 e^{\alpha t} + c_4 \alpha^2 e^{-\alpha t} & y^{(2)}(t) &= d_3 \alpha^2 e^{\alpha t} + d_4 \alpha^2 e^{-\alpha t} \\
(32) \quad x^{(3)}(t) &= c_3 \alpha^3 e^{\alpha t} - c_4 \alpha^3 e^{-\alpha t} & y^{(3)}(t) &= d_3 \alpha^3 e^{\alpha t} - d_4 \alpha^3 e^{-\alpha t}.
\end{aligned}$$

Since $\lim_{t \rightarrow \infty} y^{(1)}(t) = 0$ (because the derivative of $e^{f_\infty t} u^*(t)$ belongs to $W^{1,2}([0, \infty))$ by assumption and thus is square-integrable and continuous), we have that $d_2 = d_3 = 0$; and from $x^{(2)}(0) = 0$ (see 29) it immediately follows that $c_3 + c_4 = 0$, i.e. $c_4 = -c_3$. Using this together with the boundary condition

$$(33) \quad -\alpha^{-3} e^{f_\infty u} (x^{(3)}(u) - y^{(3)}(u)) = 1,$$

we get

$$(34) \quad -c_3 e^{f_\infty u} (e^{\alpha u} + e^{-\alpha u}) = 1 + d_4 e^{f_\infty u} e^{-\alpha u}.$$

The r.h.s. is equal to

$$(35) \quad 1 + c_3 e^{f_\infty u} (e^{\alpha u} - e^{-\alpha u}),$$

because $x^{(2)}(u) - y^{(2)}(u) = 0$ has to hold true. Thus

$$(36) \quad \begin{aligned} c_3 &= -\frac{e^{-f_\infty u} e^{-\alpha u}}{2}; \quad \text{and} \\ c_4 &= \frac{e^{-f_\infty u} e^{-\alpha u}}{2}, \quad d_4 = -e^{-f_\infty u} \sinh(\alpha u). \end{aligned}$$

Further, the boundary condition $x^{(1)}(u) - y^{(1)}(u) = 0$ implies that

$$(37) \quad c_2 = e^{-f_\infty u} \alpha.$$

Finally, we conclude from $x(u) - y(u) = 0$ that

$$(38) \quad d_1 = c_1 + e^{-f_\infty u} \alpha u.$$

Using $x(0) = 0$ (recall that u^* belongs to H_0), we get $c_1 = 0$ and consequently $d_1 = e^{-f_\infty u} \alpha u$.

Collecting all terms,

$$(39) \quad \begin{aligned} x(t) &= e^{-f_\infty u} (\alpha t - e^{-\alpha u} \sinh(\alpha t)) \quad \text{and} \\ y(t) &= e^{-f_\infty u} (\alpha u - e^{-\alpha t} \sinh(\alpha u)), \end{aligned}$$

so that u^* has the claimed form

$$(40) \quad e^{-f_\infty(t+u)} \begin{cases} \alpha t - e^{-\alpha u} \sinh(\alpha t) & \text{for } 0 \leq t \leq u; \\ \alpha u - e^{-\alpha t} \sinh(\alpha u) & \text{for } t \geq u. \end{cases}$$

□

Remarks: From (23) it follows that u^* is C^2 on $[0, \infty)$ with third derivative discontinuous at u .

The form of u^* depends on the definition of the inner product on H_0 . In this regard, Smith & Wilson (2000) do have an erroneous factor of one half in their objective function, which is corrected in this paper.

The existence of representers of the linear functionals δ_u implies that H_0 is a reproducing kernel Hilbert space with kernel function $W : [0, \infty) \times [0, \infty) \rightarrow \mathbb{R}$ given by $W(t, u) := u^*(t)$.

[For a systematic exposition on reproducing kernel Hilbert spaces, see the classical paper by Aronszajn, 1950.]

The kernel function $W(t, u)$ is exactly the Wilson function from Smith & Wilson (2000).

5. AN EXPLICIT DERIVATION OF THE SOLUTION TO THE SMITH-WILSON OPTIMIZATION PROBLEM

The following theorem states that, based on risk-free financial instruments that are not collinear (i.e., for which the matrix of payments has full rank), there exists a unique solution to the Smith-Wilson optimization problem. Furthermore, this solution has an explicit analytical expression.

Theorem 5.1. *Let $B_1, \dots, B_n$ be n risk-free financial instruments with corresponding payments c_{ji} , $i = 1, \dots, n$, at times $u_j > 0$, $j = 1, \dots, J$, and with market values v_i , $i = 1, \dots, n$. If $J \geq n$ and the matrix of payments $C := (c_{ji})$ has full rank, then the infinite-dimensional optimization problem*

$$(41) \quad \begin{array}{ll} \text{minimize} & \langle P|P \rangle \\ \text{over} & P \in \mathbf{H}_1 \end{array}$$

under the constraints

$$(42) \quad \sum_{j=1}^J c_{ji} P(u_j) = v_i, \quad i = 1, \dots, n,$$

has a unique solution $\underline{P}$ given by

$$(43) \quad \underline{P}(t) = e^{-f_\infty t} + \sum_{j=1}^J \beta_j u_j^*(t),$$

where $\beta = C(C'WC)^{-1}(v - C'd)$, $W = (w_{ij})$, $w_{ij} = u_j^(u_i)$, and $d_j = e^{-f_\infty u_j}$. In particular, $\underline{P}$ lies in the J -dimensional affine space $e^{-f_\infty t} + U$, where U denotes the linear span of $u_1^*, \dots, u_J^*$.*

Proof. The proof is based on Theorem 38.C and Theorem 43.D in Zeidler (1985). For the convenience of the reader, these fundamental results are restated in Lemma A.1 and Lemma A.2 in the appendix.

First of all, note that any $P \in \mathbf{H}_1$ can be decomposed into $P = e^{-f_\infty t} + g$ with remainder $g \in \mathbf{H}_0$ (see Section 3.1). Observe that $\langle P|P \rangle = \langle g|g \rangle$ holds true. Consequently, minimising $\langle P|P \rangle$ over the affine space $\mathbf{H}_1 = e^{-f_\infty t} + \mathbf{H}_0$ subject to the constraints $\sum_j c_{ji} P(u_j) = v_i$, $1 \leq i \leq n$, is equivalent to first minimising $\langle g|g \rangle$ over the linear space $\mathbf{H}_0$ subject to the adapted constraints

$$(44) \quad \sum_j c_{ji} (e^{-f_\infty u_j} + g(u_j)) = v_i, \quad 1 \leq i \leq n;$$

and then adding the ‘long-term’ discount curve $e^{-f_\infty t}$ to the solution.

The reason behind this reformulation is that by restricting the optimization to $\mathbf{H}_0$, we have representers u^* of the point evaluations at u at our disposal. This allows us to express the Lagrange equations as a single equation in $\mathbf{H}_0$ which can be explicitly solved.

In order to show the existence of Lagrange multipliers, we have to verify Conditions (i-iii) of Lemma A.2 in the appendix. In our context, these conditions translate into:

- (1) The optimization problem over $\mathbf{H}_0$ has a solution $\underline{g}$;
- (2) $\langle g|g \rangle : \mathbf{H}_0 \rightarrow \mathbb{R}$ is Fréchet differentiable at $\underline{g}$;
- (3) the mapping

$$(45) \quad h \mapsto d_h S(\underline{g}) := d_h \left(\sum_j c_{ji} (e^{-f_\infty u_j} + \underline{g}(u_j)) - v_i \right)_i, \mathbf{H}_0 \rightarrow \mathbb{R}^n,$$

is surjective.

For any $g, h \in \mathbf{H}_0$ such that $g \neq h$, the function

$$(46) \quad \varphi_{g,h}(t) := \langle g + t(h - g) | g + t(h - g) \rangle$$

satisfies

$$(47) \quad \varphi_{g,h}^{(2)}(t_0) = 2\langle h - g | h - g \rangle > 0$$

for $t_0 \in [0, 1]$ and is therefore strictly convex on $[0, 1]$. Hence, by Proposition 42.4 in Zeidler (1985), $\langle g|g \rangle : \mathbf{H}_0 \rightarrow \mathbb{R}$ itself is a strictly convex function. It follows from Lemma A.1 in the appendix that the optimization problem over $\mathbf{H}_0$ has a unique solution $\underline{g}$. This shows Condition 1.

Since $\mathbf{H}_0$ is a Hilbert space, it follows that $\langle g|g \rangle$ is Fréchet differentiable, with directional derivatives given by

$$(48) \quad d_h \langle g|g \rangle = 2\langle g|h \rangle$$

for all $h \in \mathbf{H}_0$. In particular, Condition 2 is satisfied.

To see that Condition 3 is satisfied, note that $d_h S(\underline{g}) = C'(h(u_j))_j$ for all $h \in \mathbf{H}_0$. Since C has rank n by the assumptions of the theorem, $C' : \mathbb{R}^J \rightarrow \mathbb{R}^n$ is surjective. So it remains to find a function $h \in \mathbf{H}_0$ that attains arbitrarily prescribed values at the points $u_1, \dots, u_J$, i.e., $h \in \mathbf{H}_0$ has to be such that $h(u_1) = a_1, \dots, h(u_J) = a_J$ for $a_1, \dots, a_J \in \mathbb{R}$ arbitrary but fixed. Of course, for each point u_j there is a compactly supported, smooth function h_j with value 0 at 0, a_j at u_j , and 0 outside some neighbourhood U_j of u_j . Choose the neighbourhoods U_j in such a way that they separate the points u_j and set $h := \sum_j h_j$. This proves Condition 3.

In view of the above, it follows from Lemma A.2 in the appendix that there are Lagrange multipliers $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ such that

$$(49) \quad 2\langle \underline{g}|h \rangle - \sum_i \lambda_i \sum_j c_{ji} h(u_j) = 0$$

holds true for all $h \in \mathbf{H}_0$. Since by Theorem 4.1 the space $\mathbf{H}_0$ is a reproducing kernel Hilbert space, we may reformulate (49) in terms of representer and thus get the following equation in $\mathbf{H}_0$:

$$(50) \quad 2\underline{g}(t) - \sum_i \lambda_i \sum_j c_{ji} u_j^*(t) = 0$$

for all $t \in [0, \infty)$. Evaluating (50) at $u_1, \dots, u_J$ results in the system of equations

$$(51) \quad \begin{aligned} 2\underline{g}(u_1) - \sum_i \lambda_i \sum_j c_{ji} u_j^*(u_1) &= 0 \\ &\vdots \\ 2\underline{g}(u_J) - \sum_i \lambda_i \sum_j c_{ji} u_j^*(u_J) &= 0 \end{aligned}$$

We introduce the notation $W = (w_{ij})$, where $w_{ij} = u_j^*(u_i)$. In order to solve for $\lambda_1, \dots, \lambda_n$, rewrite (51) in matrix form as

$$(52) \quad 2(\underline{g}(u_j))_{J \times 1} - W_{J \times J} C_{J \times n} \lambda_{n \times 1} = 0.$$

Multiplying with the transpose C' of C from the left and noting that $C'WC$ is invertible (W is positive definite and C has rank n by the assumptions of the theorem) and solving for λ gives

$$(53) \quad \lambda = 2(C'WC)^{-1}C'(\underline{g}(u_j))_j.$$

Using that $\underline{g}(u_j) = \underline{P}(u_j) - d_j$, where $d_j := e^{-f_\infty u_j}$, and plugging in the constraints $\sum_j c_{ji} \underline{P}(u_j) = v_i$ into (53) yields

$$(54) \quad \lambda = 2(C'WC)^{-1}(v - C'd).$$

Plugging this expression for λ back into (50) and adding the function $e^{-f_\infty t}$, we get the following explicit shape of the solution to the optimization problem stated in the theorem:

$$(55) \quad \begin{aligned} \underline{P}(t) &= e^{-f_\infty t} + \underline{g}(t) \\ &= e^{-f_\infty t} + \frac{1}{2}(u_j^*(t))'_j C \lambda \\ &= e^{-f_\infty t} + (u_j^*(t))'_j C (C'WC)^{-1}(v - C'd). \end{aligned}$$

This completes the proof. □

Remarks: Since $\mathbf{H}_0$ is complete, we can use the full abstract power of Lemma A.1 in the appendix to infer the existence as well as uniqueness of the solution.

The discount curve $\underline{P}(t)$ is exactly the same as the one stated in EIOPA-BoS-15/035 (2016).

In their proof, Smith & Wilson (2000) write down the solution of the optimization problem first and, in a second step, show that it satisfies all constraints and that the objective function attains a higher value for any other discount function satisfying the side conditions. For the reader this is unsatisfactory insofar as it remains completely unclear how to obtain the solution in the first place.

It is not necessary to require $P(t)$ to be decreasing. In fact, without any restriction, Theorem 5.1 can be applied in a negative interest rate environment, where discount curves need not be decreasing anymore.

The functions $u_1^*, \dots, u_J^*$ are indeed linearly independent in $\mathbf{H}_0$. Suppose, on the contrary, that $u_1^*, \dots, u_J^*$ are linearly dependent. Then there are $c_1, \dots, c_J \in \mathbb{R}$, not all 0, such that $c_1 u_1^* + \dots + c_J u_J^* = 0$. Suppose $c_j \neq 0$ and construct a smooth, compactly supported function g such that $g(u_j) = 1$ and equals 0 on the remaining points $u_1, \dots, u_{j-1}, u_{j+1}, \dots, u_J$ and at $t = 0$. Then

$$(56) \quad 0 = c_1 \langle u_1^* | g \rangle + \dots + c_J \langle u_J^* | g \rangle = c_1 g(u_1) + \dots + c_J g(u_J) = c_j \neq 0,$$

a contradiction, so that $u_1^*, \dots, u_J^*$ are linearly independent.

Knowing the shape of $\underline{P}(t)$, we can compute the implied ultimate forward rate $\underline{f}_\infty$ and check that it coincides with f_∞ . In fact, we have for $t \geq \max_j u_j$:

$$\begin{aligned} \underline{f}(t) &= -\frac{d}{dt} \log \underline{P}(t) \\ &= -\frac{d}{dt} \log \left(e^{-f_\infty t} + \frac{1}{2} (u_j^*(t))'_j C \lambda \right) \\ &= -\frac{d}{dt} \log e^{-f_\infty t} - \frac{d}{dt} \log \left(1 + \frac{1}{2} (e^{-f_\infty u_j} (\alpha u_j - e^{-\alpha t} \sinh(\alpha u_j)))'_j C \lambda \right) \\ (57) \quad &= f_\infty - \frac{\frac{\alpha}{2} e^{-\alpha t} (e^{-f_\infty u_j} \sinh(\alpha u_j))'_j C \lambda}{1 + \frac{1}{2} (e^{-f_\infty u_j} (\alpha u_j - e^{-\alpha t} \sinh(\alpha u_j)))'_j C \lambda}, \end{aligned}$$

so that

$$(58) \quad \underline{f}_\infty = \lim_{t \rightarrow \infty} \underline{f}(t) = f_\infty.$$

This is the desired relation.

From (57) we also infer that the speed of convergence to the ultimate forward rate is

$$(59) \quad |\underline{f}(t) - f_\infty| = O(e^{-\alpha t}).$$

In particular, the parameter α determines the speed of convergence.

The continuously compounded spot rate $\underline{R}(t)$ also converges to the ultimate forward rate f_∞ , which follows from

$$\begin{aligned} \underline{R}(t) &= -\frac{\log \underline{P}(t)}{t} \\ (60) \quad &= f_\infty - \frac{1}{t} \log \left(1 + \frac{1}{2} (e^{-f_\infty u_j} (\alpha u_j - e^{-\alpha t} \sinh(\alpha u_j)))'_j C \lambda \right). \end{aligned}$$

As for the ‘smoothness’ of the various interest rate curves, we conclude from the fact that the representers of the point evaluation functionals are C^2 with discontinuous third derivative (see Section 4) that the discount curve $\underline{P}(t)$ is C^2 with discontinuous third derivative, the spot

curve $\underline{R}(t)$ is C^2 with discontinuous third derivative, and the forward curve $\underline{f}(t)$ is C^1 with discontinuous second derivative. In each case, the discontinuities lie precisely at $u_1, \dots, u_J$.

The restricted ‘smoothness’ of the discount curve and, as a consequence, of the spot and forward rate curves is a direct consequence of imposing the side conditions stated in (42).

APPENDIX

In the following, we state fundamental results from Zeidler (1985) on the existence and uniqueness of optimization problems and on solving such problems with the help of Lagrange multipliers. These results cover the infinite-dimensional case.

Lemma A.1 (Existence and uniqueness of optimizers). *Let $Q : H \rightarrow \mathbb{R}$ be a function on some Hilbert space H . Let C be a non-empty, convex, closed subset of H . Then:*

- (1) *If Q is convex, continuous and unbounded, it has a minimum in C . The set of minimal points in C is non-empty, closed, convex, and bounded.*
- (2) *If, in addition, Q is strictly convex, the minimum is unique.*

Proof. Part 1 follows from Proposition 38.15 in Zeidler (1985), upon noting that any Hilbert space is a reflexive Banach space. In view of Part 1, Part 2 is a direct consequence of Theorem 38.C in Zeidler (1985). $\square$

Lemma A.2 (Existence of Lagrange multipliers). *Let $Q : E \rightarrow \mathbb{R}$ be a function on some Banach space E and let $S : E \rightarrow F$ be a function on E with values in some Banach space F . If*

- (1) *$\underline{x}$ is a minimizer of Q subject to the side condition $S(\underline{x}) = 0$;*
- (2) *Q is Fréchet differentiable at $\underline{x}$;*
- (3) *S is continuously Fréchet differentiable in a neighbourhood of $\underline{x}$ and the mapping $y \mapsto d_y S(\underline{x}), E \rightarrow F$, is surjective;*

then there is a Lagrange multiplier $\lambda \in F^$, the space of continuous linear functionals on F , such that*

$$(61) \quad d_y Q(\underline{x}) - \lambda(d_y S(\underline{x})) = 0$$

holds true for all directions $y \in E$.

For the proof see Theorem 43.D in Zeidler (1985).

Proposition A.3. Let Ω be a non-empty open subset of $\mathbb{R}$. Then $W^{1,2}(\Omega)$ is a Hilbert space.

Proof. Let g_n be a Cauchy sequence in $W^{1,2}(\Omega)$. Then, g_n as well as (any selection of) its weak derivative $D_w g_n$ are Cauchy sequences in $\mathcal{L}^2(\Omega)$. Since $\mathcal{L}^2(\Omega)$ is complete, there are $g, h \in \mathcal{L}^2(\Omega)$ such that $\|g_n - g\|_2$ as well as $\|D_w g_n - h\|_2$ converge to 0. Now, for any compactly supported

$\varphi \in C^\infty(\Omega)$, we have that

$$\begin{aligned} \langle h|\varphi \rangle_2 &= \lim_{n \rightarrow \infty} \langle D_w g_n |\varphi \rangle_2 = - \lim_{n \rightarrow \infty} \langle g_n |\varphi^{(1)} \rangle_2 \\ (62) \qquad &= - \langle g |\varphi^{(1)} \rangle_2. \end{aligned}$$

Hence, g is weakly differentiable with weak derivative given by $D_w g = h$. Since g is, up to a null set, equal to a bounded, uniformly continuous function on Ω , we conclude that $W^{1,2}(\Omega)$ is complete. $\square$

Proposition A.4. The following assertions hold true:

- (1) For any non-empty open interval of $\mathbb{R}$, the extension mapping $g \mapsto \bar{g} : W^{1,2}(\Omega) \rightarrow W^{1,2}(\bar{\Omega})$ is a linear isomorphism. In particular, $W^{1,2}(\bar{\Omega})$ is a Hilbert space.
- (2) The set of all $\varphi \in C^\infty([0, \infty))$ with compact support lies dense in $W^{1,2}([0, \infty))$.
- (3) The set of all $h \in H_0 \cap C^\infty([0, \infty))$ with compact support lies dense in H_0 .

Proof. Part 1 is clear by the definition of $W^{1,2}(\bar{\Omega})$ in Section 2.3.

As for Part 2, first note that the set of all $\varphi \in C^\infty([0, \infty))$ with compact support coincides with the set of restrictions to $[0, \infty)$ of compactly supported, smooth functions on $\mathbb{R}$. In view of Part 1, Part 2 follows from Theorem 3.18 in Adams (1975), upon noting that the interval $\Omega = (0, \infty)$ satisfies the segment property defined in Section 3.17 *ibid*.

Part 3: First of all, let $\psi \in C^\infty([0, \infty))$ be a function with compact support that satisfies

- $0 \leq \psi(t) \leq 1$;
- $\psi(t) = 1$ for $0 \leq t \leq 1$;
- $\psi(t) = 0$ for $t \geq 2$;

and the first and second derivatives of which are bounded by $0 < M < \infty$. For $0 < \delta < 1$ define $\psi_\delta(t) := \psi(\delta t)$. Then $\psi_\delta \in C^\infty([0, \infty))$ has compact support and satisfies

- $0 \leq \psi_\delta(t) \leq 1$;
- $\psi_\delta(t) = 1$ for $0 \leq t \leq \delta^{-1}$;
- $\psi(t) = 0$ for $t \geq 2\delta^{-1}$;

and the first and second derivatives of ψ_δ are bounded by δM and $\delta^2 M$, respectively.

Now, in order to prove Part 3, let $g \in H_0$ and $\varepsilon > 0$. By Part 2 of the proposition there is a $\varphi \in C^\infty([0, \infty))$ with compact support contained in $[0, N]$, $0 < N < \infty$, such that

$$(63) \qquad \sqrt{\max(\alpha^{-1}, \alpha^{-3})} \left\| \frac{d}{dt} [e^{f^\infty t} g(t)] - \varphi \right\|_{1,2} < \frac{\varepsilon}{2}.$$

Fix ψ_δ , where

$$(64) \qquad 0 < \delta < \min \left(1, \frac{1}{N}, \frac{\varepsilon^2}{16 \max(\alpha^{-1}, \alpha^{-3}) M^2 \left(\int_{[0, \infty)} |\varphi(s)| ds \right)^2} \right).$$

Letting

$$(65) \quad h(t) := e^{-f_\infty t} \int_{[0,t]} \varphi(s) ds$$

and using the properties of ψ_δ (note, in particular, that $\delta < N^{-1}$ implies that ψ_δ is equal to 1 on the support of φ), we get

$$(66) \quad \begin{aligned} \|g - \psi_\delta h\| &\leq \|g - h\| + \|(1 - \psi_\delta)h\| \\ &< \frac{\varepsilon}{2} + \left[\alpha^{-1} \int_0^{2\delta^{-1}} \left(\frac{d}{dt} \psi_\delta(t) \int_{[0,\infty)} |\varphi(s)| ds \right)^2 dt \right. \\ &\quad \left. + \alpha^{-3} \int_0^{2\delta^{-1}} \left(\frac{d^2}{dt^2} \psi_\delta(t) \int_{[0,\infty)} |\varphi(s)| ds \right)^2 dt \right]^{\frac{1}{2}} \\ &< \varepsilon. \end{aligned}$$

As $\psi_\delta h \in \mathbf{H}_0$, this proves Part 3. □

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